

DOBERLUG KIRCHHAIN, GermanyWMO: **104900**Lat: **51.6450N**Lon: **13.5747E**Elev: **321**StdP: **14.53**Time Zone: **1.00 (EUC)**Period: **82-92**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.5	11.8	-1.8	5.1	5.1	5.6	7.4	13.1	30.8	42.9	27.0	39.7	4.5	200	0.597

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	85.4	66.6	81.9	64.8	78.9	63.9	68.4	81.5	66.8	78.6	65.1	76.0	8.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.0	90.3	73.5	62.2	84.7	71.3	60.6	80.0	70.1	32.8	81.4	31.5	78.9	30.2	76.0	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.6	21.3	18.8	3.6	93.0	12.0	4.1	-5.1	96.0	-12.1	98.4	-18.8	100.7	-27.6	103.6		
(5)			2.6	72.8	11.9	1.6	-6.0	73.9	-12.9	74.8	-19.5	75.7	-28.2	76.9	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	32.0	31.3	40.0	45.7	55.2	60.0	64.4	64.3	57.5	49.8	39.9	35.3		
	DBStd	13.74	10.66	9.78	6.97	6.56	6.32	5.52	5.10	5.89	5.28	7.15	7.01	7.52		
	HDD50	2444	558	525	314	161	26	1	0	0	3	92	306	458		
	HDD65	6333	1023	944	775	580	307	167	74	87	231	472	752	921		
	CDD50	1738	0	1	4	32	187	302	447	443	230	86	4	2		
	CDD65	151	0	0	0	0	3	18	56	65	8	1	0	0		
	CDH74	2020	0	0	0	4	111	270	770	750	109	6	0	0		
	CDH80	507	0	0	0	0	4	44	211	235	13	0	0	0		
	WSAvg	8.7	10.6	9.6	9.8	8.4	7.7	7.8	7.5	6.9	7.8	8.6	9.6	9.9		
	PrecAvg	23.7	1.9	1.5	1.7	1.3	2.2	2.3	3.0	2.6	2.0	1.6	2.0	1.9		
(15) Precipitation	PrecMax	33.7	3.2	2.8	4.5	3.6	4.4	4.8	6.8	7.7	5.7	3.1	5.0	3.5		
	PrecMin	15.2	0.2	0.4	0.4	0.0	0.6	0.3	0.7	0.6	0.2	0.4	0.0	0.8		
	PrecStd	4.4	0.8	0.7	1.1	0.7	1.1	1.1	1.5	1.8	1.1	0.8	1.1	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.9	58.1	63.9	72.9	80.0	85.9	90.3	92.7	81.9	73.5	58.7	54.7		
		MCWB	48.2	50.1	53.2	56.7	63.0	65.0	68.5	69.8	64.1	62.1	52.2	49.5		
	2%	DB	49.1	52.2	58.0	67.2	77.2	81.1	85.4	86.1	77.1	68.1	54.8	52.4		
		MCWB	45.2	46.4	49.3	54.3	60.7	63.9	66.4	67.7	62.1	58.0	50.0	47.3		
	5%	DB	46.8	47.0	53.7	63.0	73.9	77.4	81.9	81.7	72.5	63.8	52.2	49.2		
		MCWB	43.2	42.4	47.0	52.0	59.0	63.0	64.6	65.4	60.5	55.9	48.3	45.9		
	10%	DB	44.4	43.6	50.5	58.5	69.8	73.5	78.6	77.9	68.2	60.5	49.5	46.2		
		MCWB	40.9	39.5	45.1	49.5	57.3	61.1	64.0	64.0	58.8	53.8	46.2	43.6		
	0.4%	WB	48.0	50.5	53.5	58.1	65.8	68.3	71.7	71.1	66.8	62.7	53.1	50.2		
		MCDB	51.5	57.6	62.0	70.1	77.3	79.1	84.9	87.5	78.3	72.8	56.8	54.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	45.9	46.4	50.4	55.2	62.6	66.3	68.8	68.5	63.8	59.2	50.8	47.7		
		MCDB	48.9	51.5	56.8	66.0	73.0	76.9	81.8	82.7	74.2	66.4	54.2	51.3		
	5%	WB	43.7	42.8	47.9	52.3	60.3	64.0	66.7	66.8	61.6	56.6	48.5	46.0		
		MCDB	46.5	46.8	52.6	61.0	70.9	74.4	78.0	79.4	70.3	61.9	51.6	49.2		
	10%	WB	41.1	39.9	45.7	50.1	58.2	61.9	64.8	64.9	59.5	54.6	46.4	43.7		
		MCDB	44.1	43.4	50.0	57.4	68.3	71.7	75.7	75.6	67.2	59.5	49.4	46.2		
	5% DB	MDBR	10.0	12.6	14.2	17.9	20.7	18.3	20.5	20.5	17.4	15.8	11.2	8.6		
		MCDBR	10.3	16.7	20.9	26.5	29.3	27.5	29.8	28.6	25.8	22.1	14.8	10.4		
		MCWBR	8.4	12.5	14.3	15.7	16.3	14.2	14.6	13.9	14.9	13.9	11.5	8.3		
		MCWBR	10.4	16.3	19.8	23.5	25.4	23.0	25.5	25.1	22.9	19.2	14.3	9.7		
(35) Mean Daily Temperature Range	5% WB	MCWBR	8.7	12.4	14.1	14.3	15.0	13.0	13.6	12.7	13.6	12.8	11.7	8.3		
	Clear-Sky Solar Irradiance	taub	0.319	0.344	0.385	0.418	0.419	0.415	0.434	0.427	0.388	0.368	0.343	0.312		
		taud	2.351	2.328	2.227	2.194	2.236	2.277	2.232	2.267	2.348	2.374	2.384	2.365		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	217	244	253	259	265	267	259	254	251	232	205	199		
		Edh at Noon	20	27	36	41	42	40	42	38	31	25	20	17		
(44) All-Sky Solar Radiation	RadStd		227	445	764	1262	1548	1650	1558	1349	974	561	266	173		
			28	77	95	163	185	134	198	117	121	66	38	19		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (49 neighbors)		N/A	N/A	N/A	N/A	+0.61	+0.68	N/A	N/A	N/A	N/A	

Nomenclature: See separate page